

Pasan Rathnayake

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Graduate Software Engineer – System & AI Agentic Developer

PROFESSIONAL SUMMARY

Graduate software engineer with experience in software development, AI-assisted workflows, and distributed systems. Experienced in C++, Python, JavaScript, SQL, Kafka, and containerised tooling. Skilled in building real-time and event-driven systems using LangGraph, LangChain, and modern APIs. Adaptable, systems-focused, and project-driven, with experience across enterprise, eCommerce, and automation-focused environments.

KEY SKILLS

Backend & Languages: Python, C++, JavaScript, Dart, SQL

Frameworks & APIs: Node.js, React, Flutter, REST

DevOps & Tools: Kafka, Docker, Git, GitHub Actions, Cloud fundamentals (AWS)

Databases: MongoDB, MySQL, Firebase

AI & Automation: LangChain, LangGraph, OpenAI API, n8n

Application Development: WordPress, Shopify

Practices: Agile, SDLC, CI/CD

Soft Skills: Problem-Solving, Critical Thinking, Leadership, Communication, Adaptability

TECHNICAL STRENGTHS SNAPSHOT

Web Development | Backend Systems | Distributed & Event-Driven Architecture | Automation Workflows | Secure Coding | Threat-aware system design | Prompt Engineering | Local LLM tooling

DEVELOPER TOOL STACK

Postman | Wireshark | Metasploit | Kali Linux | VS Code | Docker Desktop | WSL | Linux | Claude Code | Codex | Antigravity | OpenCode

ADDITIONAL TECHNICAL CAPABILITIES

- Debugging and maintaining large existing codebases
- API development and backend optimisation
- Containerised development and deployment workflows
- CI/CD pipelines and version control best practices
- Designing and maintaining scalable, maintainable system architectures

TECHNICAL & AI PROJECTS

Plan + Approve Web-Search AI Agent — *Python, LangGraph, LangChain, OpenAI API*

- Built a CLI-based AI agent capable of generating plans, requesting human approval, and performing web searches
- Implemented interrupt/resume control, summarised memory, and structured JSON outputs

Uber-style Backend Framework — *C++, gRPC, Kafka, RabbitMQ*

- Developed a modular backend framework inspired by large-scale ride-sharing systems
- Implemented multithreaded microservices, centralised logging, message queues, and geospatial indexing

Instagram-style Mobile Application — *React Native, Node.js, MongoDB*

- Built a full-featured mobile application with user profiles, image upload and deletion, and post interaction
- Used Redux for state management and implemented backend APIs for core functionality

eHealth Medical System — *React, Node.js, Flutter, Docker*

- Developed a full-stack platform for physiotherapists and clients to manage appointments and reports
- Followed SDLC best practices and implemented unit testing

PROFESSIONAL EXPERIENCE

AI Agentic Developer Intern

Thiru Labs | Feb 2026 – Current

- Implement prototype for an AI first Harness, for full automated protection from cyber threats.
- Contribute to developing parental monitoring mobile applications.

Information Technology Analyst

Mentone Furniture Clearance | Sep 2025 – Dec 2025

- Supported backend systems and technical operations for eCommerce platforms
- Developed and maintained custom features using WordPress and Shopify
- Provided technical troubleshooting and backend improvements
- Improved maintainability and system stability through structured enhancements

Technician & System Integrator

AVI Solutions | Dec 2023 – Aug 2025

- Installed and configured IoT networks and automation systems
- Diagnosed and resolved hardware, software, and connectivity issues
- Collaborated with engineering teams to improve system reliability
- Documented internal processes to support onboarding and maintenance

Software Engineer Intern

London Stock Exchange Group | Dec 2022 – Jun 2023

- Developed backend components using C++ and Python
- Worked with Kafka-based stream processing systems
- Participated in Agile ceremonies and technical design reviews
- Contributed to performance and throughput improvements in distributed architectures

EDUCATION

Master of Information Technology (Specialised in Cyber Security)

Deakin University, Australia

Graduated: August 2025

Bachelor of Computer Science (Major in Software Engineering)

Edith Cowan University, Australia

Graduated: March 2023

Certificate in Computer Applications

University of Colombo (IHRA), Sri Lanka

Graduated: March 2020

ADDITIONAL INFORMATION

Availability: Immediate

References: Available on request